

EXPERIMENT 4:

Implement a finite-state machine for morphological parsing. In this example, we'll create a simple machine to generate plural forms of English nouns using python.

AIM:

To implement a Finite State Machine (FSM) for morphological parsing by generating plural forms of English nouns using Python.

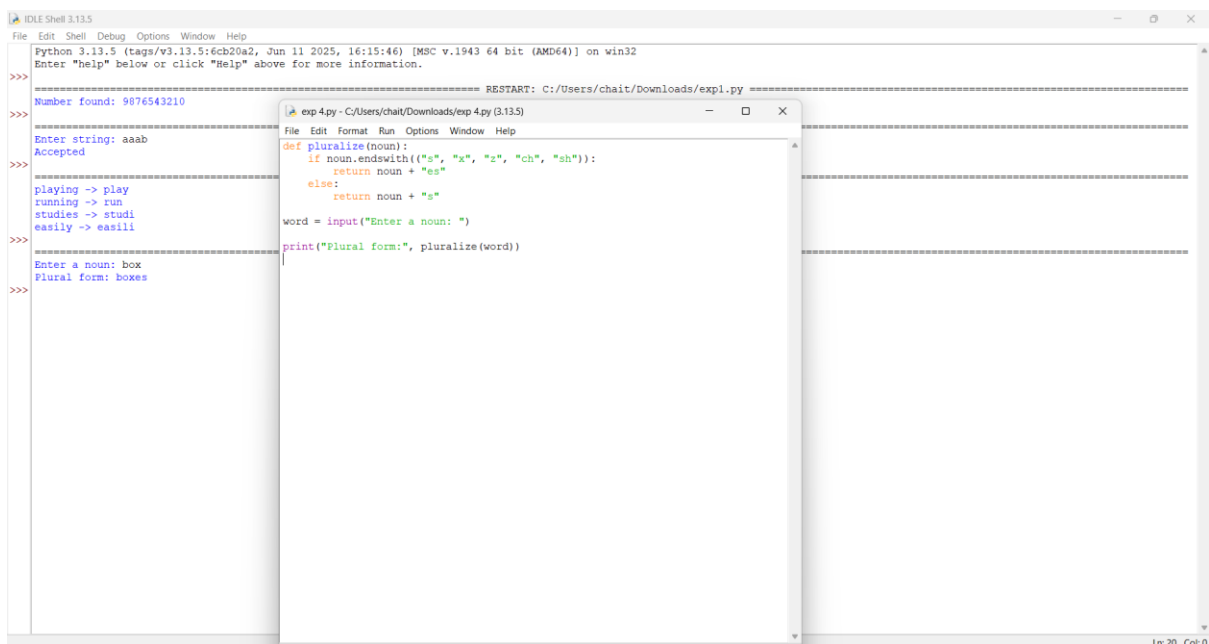
PROGRAM:

```
def pluralize(noun):  
    if noun.endswith(("s", "x", "z", "ch", "sh")):  
        return noun + "es"  
    else:  
        return noun + "s"
```

```
word = input("Enter a noun: ")
```

```
print("Plural form:", pluralize(word))
```

OUTPUT:



The screenshot shows a Python IDE with two windows. The background window is a terminal showing the execution of a program. The foreground window is a code editor showing the source code of the program. The code defines a function `pluralize(noun)` that checks if a noun ends with "s", "x", "z", "ch", or "sh". If it does, it appends "es"; otherwise, it appends "s". The program prompts the user to enter a noun and prints the plural form. The terminal output shows the program running successfully, with the input "box" resulting in the plural form "boxes".

```
Python 3.13.5 (tags/v3.13.5:6cb20a2, Jun 11 2025, 16:15:46) [MSC v.1943 64 bit (AMD64)] on win32  
Enter "help" below or click "Help" above for more information.  
>>>  
Number found: 9876543210  
===== RESTART: C:/Users/chaith/Downloads/exp1.py =====  
>>>  
Enter string: aaab  
Accepted  
>>>  
playing -> play  
running -> run  
studies -> studi  
easily -> easilli  
===== RESTART: C:/Users/chaith/Downloads/exp1.py =====  
>>>  
Enter a noun: box  
Plural form: boxes  
>>>
```

```
exp 4.py - C:/Users/chaith/Downloads/exp 4.py (3.13.5)  
File Edit Format Run Options Window Help  
def pluralize(noun):  
    if noun.endswith(("s", "x", "z", "ch", "sh")):  
        return noun + "es"  
    else:  
        return noun + "s"  
word = input("Enter a noun: ")  
print("Plural form:", pluralize(word))
```